Neuromorphic ASR
Transducer Models on Memristor Hardware

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Outline

Related Works

Motivation

Prerequisites

Experiments and Results

Future Work

Related Works

- **Monotonic Recurrent Neural Network Transducer (RNN-T) and Decoding Strategies** [Tripathi$^+$ 19]
  - Introduces monotonic alignments for RNN-T
- **Conformer: Convolution-Augmented Transformer for Speech Recognition** [Gulati$^+$ 20]
  - Introduces the Conformer encoder
- **Synaptogen: A Cross-Domain Generative Device Model for Large-Scale Neuromorphic Circuit Design** [Hennen$^+$ 24]
  - Synaptogen: Memristor device model for a family of devices
- **SynaptogenML** [Rossenbach$^+$ 25]
  - Extends Synaptogen and introduces SynaptogenML

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- Memristor hardware enables fast, energy-efficient neural network inference.
  → but introduces analog inaccuracies.
- CTC-based Conformers have been evaluated on memristor hardware [Rossenbach^{+} 25].
- Transducers add two components, which are repeatedly invoked at inference:
  - Predictor network
  - Joiner network
- Objective:
  - Characterize the effect of quantization and device noise on transducer ASR.
  - Assess training and decoding countermeasures.

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Overview

- A memristor is an analog computing device with a programmable resistance state [Strukov+ 08].
- Enables fast, energy-efficient implementation of Vector-Matrix Multiplication (VMM)

  → and thus neural networks.
- Limitations:
  - Introduces analog inaccuracies.
  - Production is still in early stages.
  - Experiments are simulated using SynaptogenML [Rossenbach+ 25].

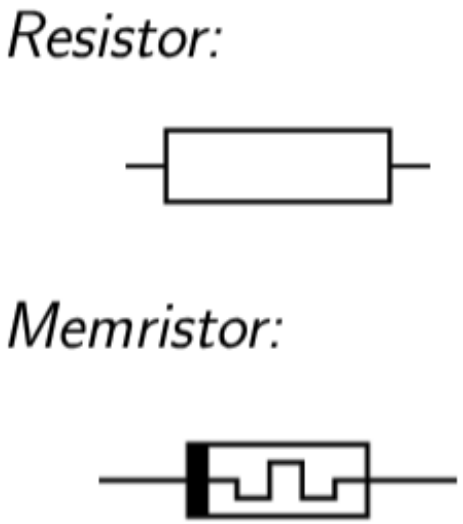


Figure: Resistor and memristor schematics

Analog In-Memory Computing

- Multiplication computation via Ohm's law:

$$\boldsymbol{V = IR} \tag{1}$$

- Voltage $\boldsymbol{V}$, current $\boldsymbol{I}$, and resistance $\boldsymbol{R}$.
- Possible to rewrite resistance as conductance $\boldsymbol{G}$:

$$\boldsymbol{I = GV} \quad \text{with} \quad \boldsymbol{G = \frac{1}{R}} \tag{2}$$

- Addition done via Kirchhoff's law:
  - For any connected node in a circuit, with $\boldsymbol{N}$ connected lines:

$$\sum_{n=0}^{N} \boldsymbol{I_n = 0} \;\rightarrow\; \boldsymbol{I_0 = -\sum_{n=1}^{N} I_n} \tag{3}$$

- Currents can be trivially added by simply connecting electric lines.

Analog In-Memory Computing

- Resistors as the connecting element between vertical and horizontal lines.
- Each resistor's conductance is an element of a constant matrix.
- Column sum via line connection.

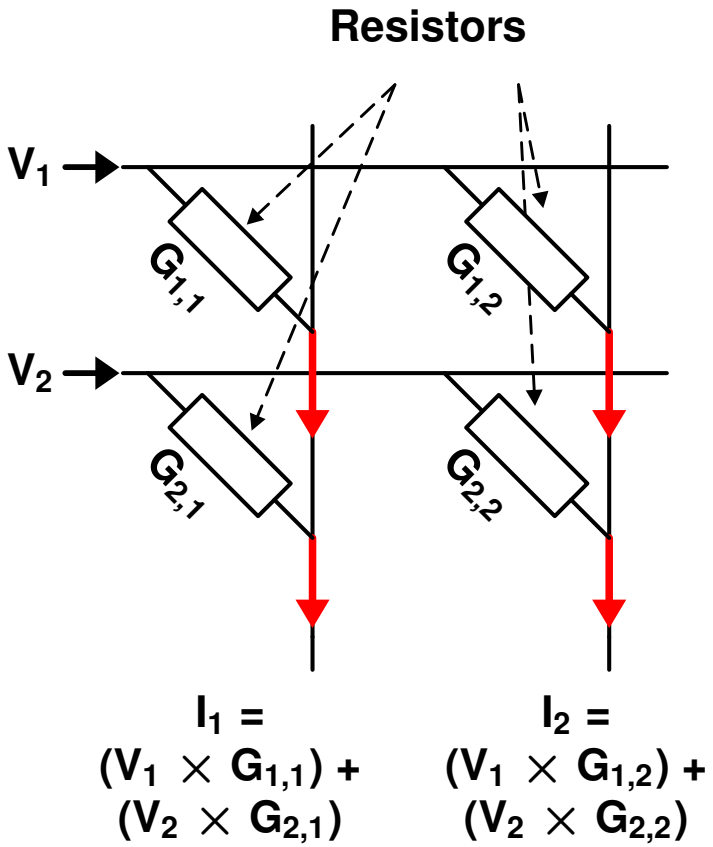


Figure: Crossbar array performing analog VMM [Rossenbach+ 25]

The Memristor

Key Idea: Encode each weight bit as a high- or low-resistance state.

- Let V_S be the set voltage and V_R the reset voltage thresholds.
- Apply $V < V_S \rightarrow$ set to low resistance.
- Apply $V > V_R \rightarrow$ set to high resistance.
- Intermediate states are difficult to program reliably for the devices measured in Synaptogen.
- Memristor state is non-volatile.

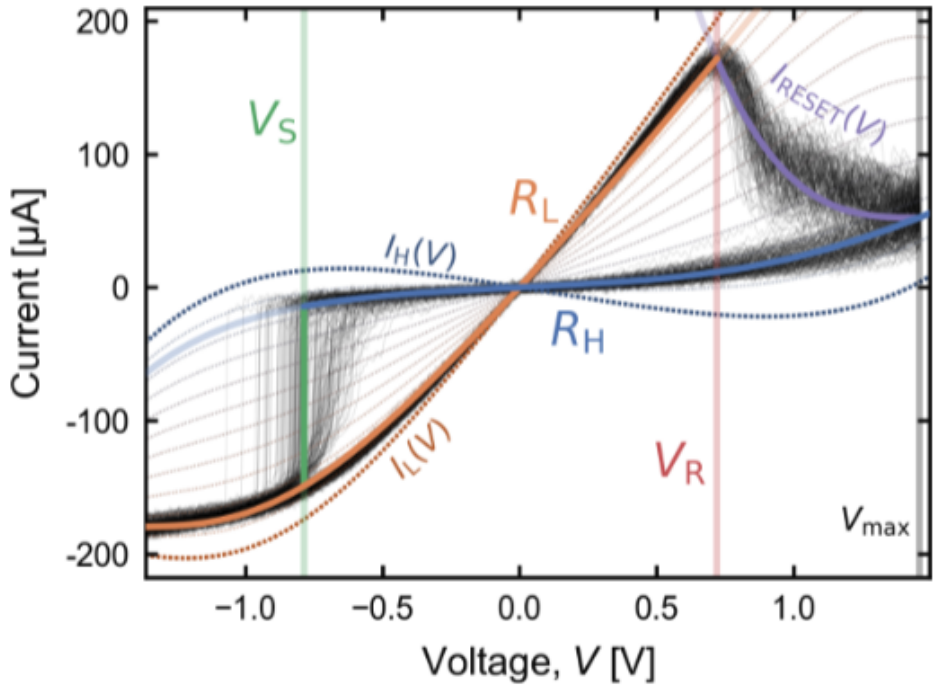


Figure: I–V characteristic curve of a memristor [Hennen+ 24]

R_L/R_H: low-/high-resistance state; $I_L(V)/I_H(V)$: corresponding current; V_{max}: maximum applied voltage; $\mathrm{RESET}(V)$: current during the reset transition.

Limitations So Far

- Resulting resistance state varies with each programming cycle.
- Non-linear relation of current and voltage.

- Only positive, non-zero weights possible.
- Current leakage prohibits a true zero.

Memristor Crossbar Layout

- Use positive and negative bitlines to create memristor pairs
- Each row-wise pair of memristors represents a weight

→ For N input rows, with row index i and output-column index j:

$$I_j = \sum_{i=1}^{N} V_i G_{i,j}^{+} - \sum_{i=1}^{N} V_i G_{i,j}^{-}$$

Benefits and limitations:

- Cannot set paired memristors to identical conductances

→ Difference is usually close to zero, but not guaranteed.

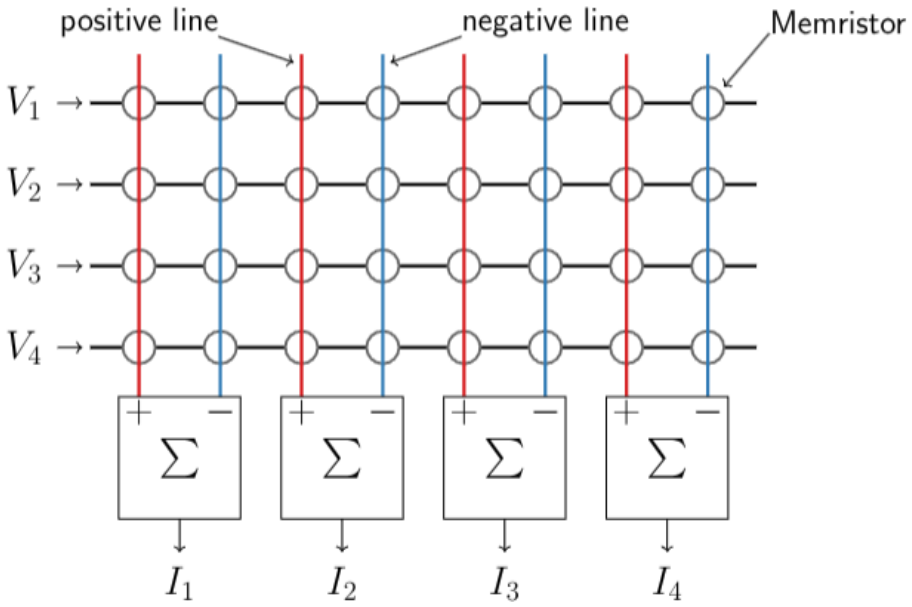


Figure: Paired memristor array structure [Aguirre+ 24]

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Transducer Deployment Scenarios

- Which components of the model can be deployed on the device?
  - ▷ Encoder only vs. encoder + predictor vs. encoder + predictor + joiner.
- Tradeoff: more offloading ⇒ more energy efficiency, but also more noise.
- Optional external language model during decoding.

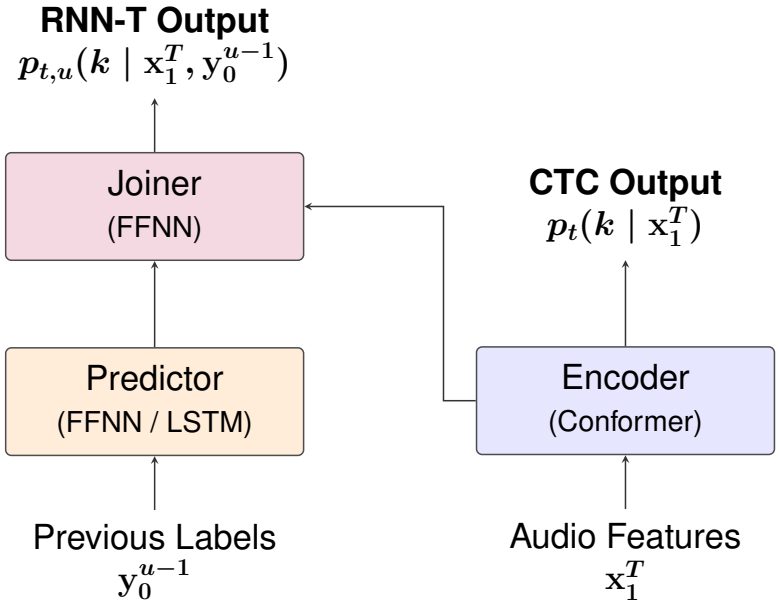


Figure: CTC vs. Transducer architecture components

FFNN: feed-forward neural network; **LSTM**: long short-term memory; **LM**: language model.

k: candidate label or blank; t: acoustic-frame index; u: output-label index.

$\mathrm{x}_1^T = (\mathrm{x}_1, \ldots, \mathrm{x}_T)$: audio-feature sequence; $\mathrm{y}_0^{u-1} = (y_0, \ldots, y_{u-1})$: previously emitted labels;

Experimental Setup: Models and Quantization

- Model types:
  1. Connectionist Temporal Classification (CTC) [Graves+ 06]: encoder only.
  2. FFNN transducer [Graves 12]: 2-layer, 640-dim feed-forward predictor.
     - Conditions on the previous-label embedding.
  3. Full-context transducer [Graves 12]: LSTM predictor over label history.
- Both transducers use monotonic alignments [Tripathi+ 19].
- Model sizes (million parameters): CTC `TBD`; FFNN transducer `TBD`; full-context `TBD`.
- Quantization: Quantization-Aware Training (QAT).
  - 8-/4-bit weights, 8-bit activations; zero-point 0.
  - Per-tensor symmetric quantization.

Experimental Setup: Training

- Data: LibriSpeech [Panayotov$^+$ 15], 960 h. 184 Byte-Pair Encoding (BPE) subword labels.
- Features: 80 log-mel filters, 10 ms frame shift; Downsampling factor 4, SpecAugment during training;
- Encoder: 12-layer Conformer [Gulati$^+$ 20], 512-dim.
- Training: 100 epochs, RAdam optimizer, ϵ=1e-12, Weight Decay=0.01
  - OCLR: initial LR 7×10^{-6}; peak 5×10^{-4} after 48 epochs; decay to 5×10^{-5} over 48 epochs; final LR 1×10^{-7} over 4 epochs.
- Runs:
  - Full-precision baseline and full pretraining under QAT.
  - QAT models simulated on-device for 3 (transducer) or 5 (CTC) runs. Results reported as mean $\pm$ standard deviation.
- Results reported on LibriSpeech dev-other, for lex-free (greedy) decoding and tree search with/without 4-gram LM.

CTC Results

Table: WER (%). 8 bit and 4 bit quantization vs. baseline. vs. memristor simulation.

		Lex-Free	Tree	
Evaluation	**Weight bits**	**No LM**	**No LM**	**4-gram**
Baseline	FP32	7.1	6.5	5.2
QAT	8	7.3	6.8	5.3
QAT	4	8.3	7.6	5.7
Memristor simulation	8	13.5 ± 0.37	10.4 ± 0.20	6.7 ± 0.09
Memristor simulation	4	11.9 ± 0.18	10.1 ± 0.12	6.8 ± 0.12

FFNN Transducer Results: 8-bit Quantization

Table: WER (%; memristor mean ± standard deviation). 8-bit quantization vs. baseline. vs. memristor simulation.

Modules Quantized			Lex-Free	Tree	
Encoder	**Predictor**	**Joiner**	**No LM**	**No LM**	**4-gram**
Baseline					
–	–	–	5.9	5.8	4.8
QAT				on LibriSpeech dev-other:	
✓	–	–	6.3	6.1	4.9
✓	✓	–	6.4	6.2	5.0
✓	✓	✓	6.4	6.2	4.9
Memristor Simulation					
✓	–	–	9.8 ± 0.06	8.7 ± 0.03	6.7 ± 0.07
✓	✓	–	10.7 ± 0.10	8.9 ± 0.11	6.7 ± 0.04
✓	✓	✓	44.7 ± 1.95	56.1 ± 10.90	18.8 ± 1.24

FFNN Transducer Results: 4-bit Quantization

Table: WER (%; memristor mean ± standard deviation). 4-bit quantization vs. baseline. vs. memristor simulation.

Modules Quantized			Lex-Free	Tree	
Encoder	**Predictor**	**Joiner**	**No LM**	**No LM**	**4-gram**
Baseline					
–	–	–	5.9	5.8	4.8
QAT					
✓	–	–	7.2	6.8	5.2
✓	✓	–	7.3	6.9	5.3
✓	✓	✓	8.2	7.8	5.7
Memristor Simulation					
✓	–	–	10.1 ± 0.04	9.1 ± 0.12	6.6 ± 0.07
✓	✓	–	10.4 ± 0.11	9.3 ± 0.12	6.7 ± 0.06
✓	✓	✓	37.4 ± 1.33	29.6 ± 2.39	12.8 ± 0.72

FFNN Transducer: Error Breakdown

Table: Full model FFNN Transducer under QAT (8 weight bits) and memristor simulation.

Evaluation	Search	WER (%)	Substitutions (%)	Insertions (%)	Deletions (%)
Baseline	Lex-free	5.9	4.9	0.5	0.5
Baseline	Tree	5.7	4.6	0.8	0.3
QAT	Lex-free	6.5	5.4	0.5	0.6
QAT	Tree	6.2	5.0	0.8	0.4
Memristor simulation	Lex-free	44.7 $\pm$ 1.95	16.6 $\pm$ 0.15	0.1 $\pm$ 0.00	28.0 $\pm$ 2.51
Memristor simulation	Tree	56.1 $\pm$ 10.90	7.1 $\pm$ 1.23	0.1 $\pm$ 0.10	48.9 $\pm$ 14.61

FFNN Transducer: Internal LM Compensation

Table: WER (%) for Tree + 4-gram LM search. Noisier models benefit from higher ILM compensation.

(a) Fixed ELM scale = 0.6

	Modules Quantized		ILM Scale				
	Encoder	**Predictor**	0.1	0.2	0.3	0.4	0.5
Baseline	–	–	**4.8**	**4.8**	**4.8**	5.0	5.1
QAT	✓	–	**5.1**	**5.1**	**5.1**	5.3	5.4
	✓	✓	**5.0**	**5.0**	5.1	5.2	5.4
Memristor	✓	–	17.4 ± 1.60	8.5 ± 0.34	7.0 ± 0.06	6.8 ± 0.08	$\mathbf{6.7 \pm 0.10}$
	✓	✓	18.7 ± 1.58	8.8 ± 0.23	7.2 ± 0.09	$\mathbf{6.8 \pm 0.03}$	6.9 ± 0.10

(b) Optimal LM Scales

	Modules Quantized		Scales		
	Encoder	**Predictor**	**ILM**	**ELM**	**WER (%)**
Baseline	–	–	0.1	0.6	4.8
QAT	✓	–	0.2	0.7	5.0
	✓	✓	0.2	0.7	5.0
Memristor	✓	–	0.3	0.5	6.7 ± 0.07
	✓	✓	0.4	0.5	6.7 ± 0.04

ILM: Internal Language Model; **ELM**: External Language Model.

Full-Context Transducer Results: 8-bit Quantization

Table: WER (%; memristor mean ± standard deviation) on LibriSpeech dev-other: 8-bit quantization vs. baseline.

Modules Quantized			Lex-Free	Tree	
Encoder	**Predictor**	**Joiner**	**No LM**	**No LM**	**4-gram**
Baseline					
–	–	–	5.8	5.6	4.9
QAT					
✓	–	–	5.9	5.7	5.2
✓	✓	–	6.1	5.9	5.2
✓	✓	✓	6.1	5.9	5.2
Memristor Simulation					
✓	–	–	9.0 ± 0.29	8.1 ± 0.16	7.1 ± 0.07
✓	✓	–	12.8 ± 0.65	10.7 ± 0.84	8.0 ± 0.28
✓	✓	✓	29.5 ± 1.14	39.5 ± 1.13	TBD

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Future Work

- Fine-tuning:
  - Fine-tuning from a full-precision checkpoint for various configurations
- Error Analysis:
  - Substitutions vs. insertions vs. deletions.

  → How can we improve the degradations in the lexicon-free search?
  - Study acoustic posterior distortion under noise.
- Generalization:
  - Evaluation and training on larger/out-of-domain datasets.

Future Work: Search

- Decoder Parameters:
  - Tune emission-score penalties and biases (e.g., for blank emissions).
  - Tune softmax temperature to sharpen or soften the distribution under noise.
- Search Space vs. Model & QAT Level:
  - Larger search spaces require more energy.
  - Study the impact of restricting the search space on WER.

  → How many more hypotheses are needed for on-device (simulated) vs. 4-bit vs. 8-bit vs. full-precision runs?

  → Compare hypothesis expansion across CTC and transducer variants.
- Search Paradigms:
  - Explore alternative search algorithms and language model combinations (e.g., lexicon-free with LSTM/transformer-based LM).

Future Work: Robust Training & Countermeasures

- Noise-and-Quantization Aware Training:
  - Inject simulated crossbar conductance noise directly into training forward pass.
- Selective Layer Mapping:
  - Deal with noise in noise-sensitive components (e.g., joint network).

  → Mixed-precision quantization
- Denoising & Output Compensation:
  - Post-processing to correct analog inaccuracies.

  → Train a small neural network to denoise the memristor output.

  → Use a small calibration dataset to learn a linear transformation to correct the output.

Thank you for your attention!

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In-Memory Computing [Rossenbach+ 25]

- Various technologies exist, with varying tradeoffs:
  - Chip complexity and density
  - Energy consumption
  - Speed and reliability
- This talk focuses on memristive hardware, in particular resistive RAM (ReRAM):
  - Key benefits: low energy consumption, non-volatility

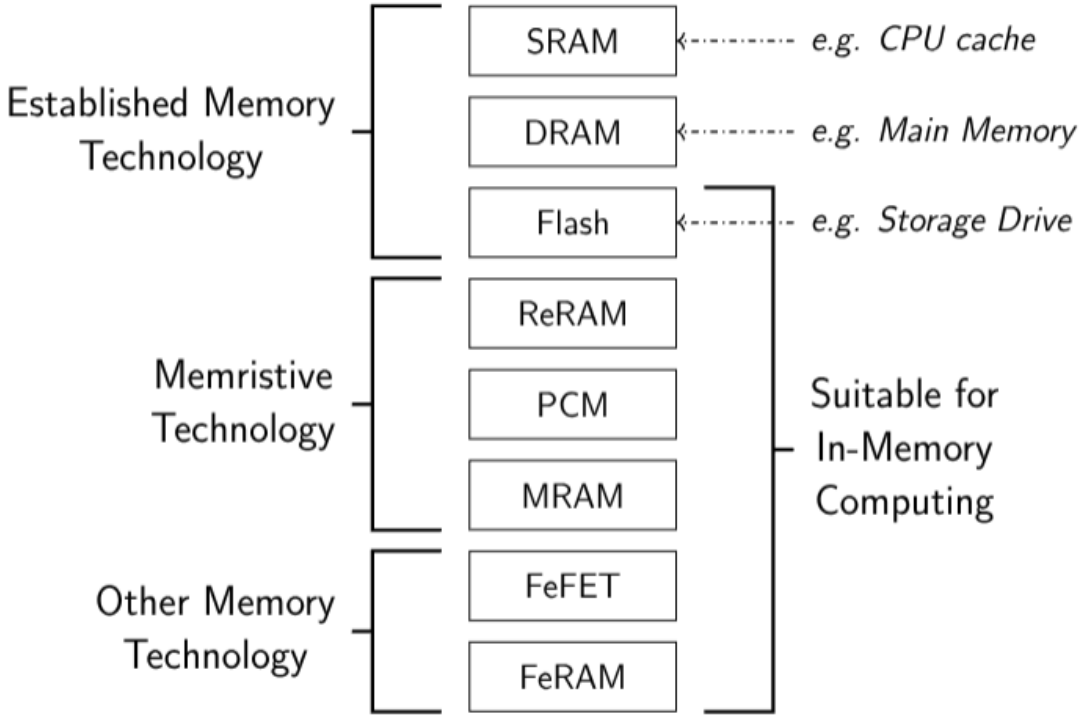


Figure: Taxonomy of In-Memory Computing technologies

Quantization

- The number of memristors required is proportional to the precision of the weights.
  - Need to reduce the precision of the weights $\rightarrow$ quantization.
- Quantization-Aware Training (QAT) performs better than Post-Training Quantization (PTQ), but requires more training time [Gholami$^+$ 21].

$\rightarrow$ High performance gap at low precision

Table: WER (%) on TED-LIUMv2 dev, Conformer CTC, 4-gram Language Model (LM) [Rossenbach$^+$ 25]

Weight bits	FP32	8	6	5	4	3	2
PTQ	7.2	7.2	7.9	8.9	11.4	30.7	–
QAT		7.3	7.7	7.4	7.8	8.3	22.1

Memristor Neural Layers [Rossenbach$^+$ 25]

- Tile the weight matrix:
  - Tile weight matrix with arbitrary dimensions into smaller fixed-size arrays.
  - E.g., 512×512 weight matrix $\rightarrow 16$ crossbar arrays of size 128×128.
- Slice weights into memristor states:
  - Binary setting (high vs. low resistance).
  - Stacking of memristors for each magnitude bit (k-bit weight requires $k - 1$ arrays).
- Pair positive $(+)$ and negative $(-)$ bitlines.

Memristor Neural Layers

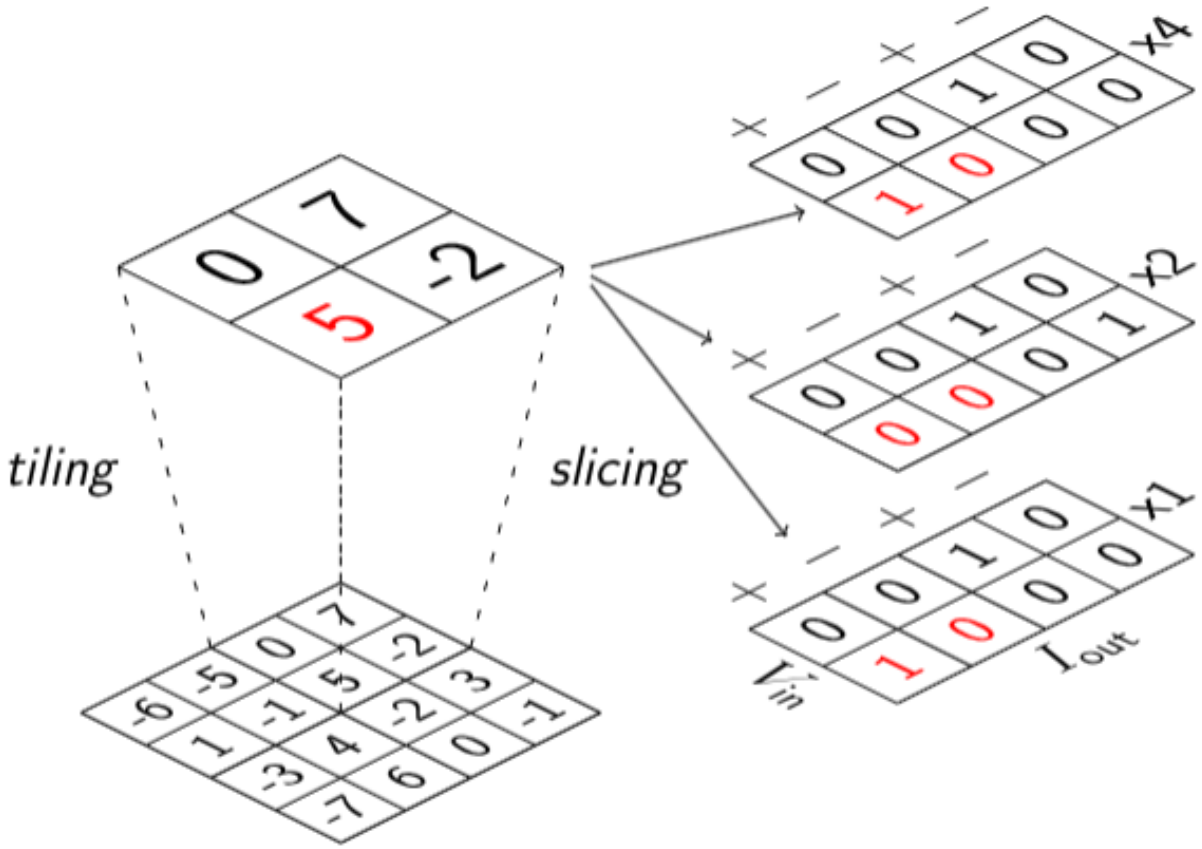


Figure: Mapping a weight matrix on memristor arrays [Rossenbach^{+} 25]

FFNN Transducer: 8-bit vs. 4-bit Quantization

Table: WER (%; memristor mean ± standard deviation) on LibriSpeech dev-other.

	Modules Quantized			Lex-Free	Tree	
Weight bits	**Encoder**	**Predictor**	**Joiner**	**No LM**	**No LM**	**4-gram**
Baseline						
–	–	–	–	5.9	5.8	4.8
QAT						
8	✓	–	–	6.3	6.1	4.9
8	✓	✓	–	6.4	6.2	5.0
8	✓	✓	✓	6.4	6.2	4.9
4	✓	–	–	7.2	6.8	5.2
4	✓	✓	–	7.3	6.9	5.3
4	✓	✓	✓	8.2	7.8	5.7
Memristor Simulation						
8	✓	–	–	9.8 ± 0.06	8.7 ± 0.03	6.7 ± 0.07
8	✓	✓	–	10.7 ± 0.10	8.9 ± 0.11	6.7 ± 0.04
8	✓	✓	✓	44.7 ± 1.95	56.1 ± 10.90	18.8 ± 1.24
4	✓	–	–	10.1 ± 0.04	9.1 ± 0.12	6.6 ± 0.07
4	✓	✓	–	10.4 ± 0.11	9.3 ± 0.12	6.7 ± 0.06
4	✓	✓	✓	37.4 ± 1.33	29.6 ± 2.39	12.8 ± 0.72

Run counts not supplied in the aggregate report. TBD: QAT-only scores not supplied.

Full QAT FFNN Transducer Deep Dive

Table: WER (%) on LibriSpeech dev-other. The entire 8-bit model is trained under QAT.

Memristor Simulated Modules			Lex-Free	Tree	
Encoder	**Predictor**	**Joiner**	**No LM**	**No LM**	**4-gram**
✓	✓	✓	44.7 ± 1.95	56.1 ± 10.90	18.8 ± 1.24
✓	–	✓	45.4 ± 4.99	55.1 ± 6.60	21.8 ± 4.84
✓	✓	–	9.8 ± 0.03	8.7 ± 0.10	6.7 ± 0.03
✓	✓	1st Layer Only	11.2 ± 0.18	9.4 ± 0.09	6.8 ± 0.05

FFNN Transducer: Fine-Tuning vs. QAT from Scratch

Table: WER (%; memristor mean ± standard deviation) on LibriSpeech dev-other: Model configuration - FFNN transducer with encoder under QAT. Memristor results use 3 simulation runs.

(a) QAT

		Lex-Free	Tree	
Training	**Weight bits**	**No LM**	**No LM**	**4-gram**
Fine-tuned	8	6.6	6.3	5.0
Full PTN	8	6.5	6.2	5.1
Fine-tuned	4	7.2	7.0	5.1
Full PTN	4	7.2	6.8	5.2

(b) Memristor Simulation

		Lex-Free	Tree	
Training	**Weight bits**	**No LM**	**No LM**	**4-gram**
Fine-tuned	8	8.1 ± 0.06	7.5 ± 0.07	5.8 ± 0.08
Full PTN	8	9.8 ± 0.06	8.7 ± 0.03	6.7 ± 0.07
Fine-tuned	4	13.1 ± 0.06	11.2 ± 0.11	7.2 ± 0.05
Full PTN	4	10.1 ± 0.04	9.1 ± 0.12	6.6 ± 0.07

- Full PTN models trained from scratch under QAT.
- Finetuned QAT models - FP32 checkpoint, finetuned for 25 epochs.
  - 4 bit models finetuned for 50 epochs, have similar performance.
- 8 bit - finetuned models perform better than full PTN.
- 4 bit full PTN models show better performance.
- Similar trends observed for various other configurations of FFNN transducer.

Device VMM Operation Statistics

operation	average	std-dev	min	max
1.0×1	1.016	0.063	0.707	1.27
0.1×1	$0.989 \cdot 10^{-1}$	$0.062 \cdot 10^{-1}$	$0.688 \cdot 10^{-1}$	$1.24 \cdot 10^{-1}$
0.01×1	$0.985 \cdot 10^{-2}$	$0.069 \cdot 10^{-2}$	$0.721 \cdot 10^{-2}$	$1.20 \cdot 10^{-2}$
1.0×0	$8.52 \cdot 10^{-4}$	$4.75 \cdot 10^{-2}$	-0.2751	0.2455
1.0×0.5	0.476	0.094	0.109	1.06

Table: Hardware measurement statistics for simulated memristor VMM operations [Rossenbach+ 25].